What is HTTP

HTTP, short for Hypertext Transfer Protocol, is used for exchanging information over the net.

- Foundation of the Web
- Makes the internet work
- Way for browsers and servers to talk forth and back
- Without it, no web pages, no URL's, no hyperlinks
- Without it, users would have to know the exact Ip address

Circles around a Request-Response Cycle

- Client sends HTTP request messages, containing info's such as requested resources and any additional parameters
- Server receives and responds with a generated message by using its resources
- Server sends it back to the client
- Client receives and processes it, typically by rendering the content in a web browser or displaying it in an app
- Client may initiate other requests and the cycle repeats

How to create HTTP Requests

URL, HTTP method, list of headers (request headers) and the request body is needed to create a valid HTTP request.

Here is an example:

```
GET /watch?v=8PoQpnlBXD0 HTTP/1.1
Host: www.youtube.com
Cookie: GPS=1; VISITOR_INFO1_LIVE=kOe2UTUyPmw; YSC=Jt6s9YVWMd4
```

1. First line specifies HTTP method, path and protocol version. In this case, GET request for the video located at the path /watch… using the HTTT/1.1 Protocol
2. The second line specifies the host of the website, which is youtube.com
3. Third line contains cookie header, used for sending + storing small pieces of data on the client side. In this case, the cookie header contains three values: GPS=1, VISITOR_INFO1_LIVE=kOe2UTUyPmw, and YSC=Jt6s9YVWMd4. These values can be used by YouTube to track and personalize the user's experience.

What is an HTTP Request URL

When a web browser attempts to access an image on the internet, sends a request to the server using an URL. This URL is unique and points to a specific resource on the server.

A resource can be anything that has a name and can be accessed with a unique identifier like a user, product, article, document, or image. You can think of resources as nouns.

What are HTTP Request Methods

HTTP METHODS	DEFINITION
HEAD	Asks the server for status (size, availability) of a resource.
GET	Asks the server to retrieve a resource.
POST	Asks the server to create a new resource.
PUT	Asks the server to edit/update an existing resource.
DELETE	Asks the server to delete a resource.

What are HTTP Request Headers

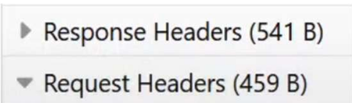
These are additional pieces of information that are sent by the client as part of an HTTP request. They have a name/value format. That is:

```
Name: Value
```

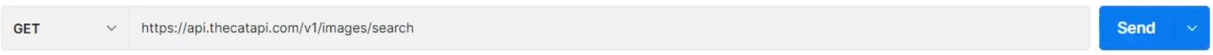
They provide context and additional instructions to the server, which can be used to process the request or customize the response.

How to check HTTP Headers

Press Ctrl+Shift+I , paste any link of your choice and check at the Network Window Headers. You should see Response as well as Request Headers like this:



You can also check them with Postman:



Here is an example of an HTTP request header:

```
GET /api/data HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko
Accept: application/json
Accept-Language: en-US,en;q=0.5
Authorization: Token abc123
Cache-Control: no-cache
Connection: keep-alive
Referer: https://www.google.com/
Pragma: no-cache
```

The request includes ten headers

Headers	Definition
Host	This header specifies the hostname of the server that the client is trying to connect to.
User-Agent	This header provides information about the client that is making the request (in this case, a version of the Chrome browser).
Accept	This header specifies the MIME types of data that the client is willing to accept in the response.
Accept-Language	This header specifies the preferred language(s) for the response.
Authorization	This header provides an access token (in this case, Token abc123) for authentication purposes.
Cache-Control	This header specifies caching directives for both requests and responses.
Connection	This header specifies options for the connection handling between client and server.
Referer	This header specifies the URL of the page that linked to the current page.

Headers	Definition
Pragma	This header specifies implementation-specific directives that might apply to any agent along the request-response chain.
Content-Type	This header specifies the MIME type of the data that is being sent in the body of the request, but it is not used in this example because this is a GET request without a request body.

What is an HTTP Request Body

The request body is the data that is sent from the client to the server as part of an HTTP request. The example below shows how to upload an image to the Cat API server

First you need an API Key, by signing up. Click on the Homepage, go to documentation, to get access to the entire API spec. Paste the link in Postman, after that you can check all the headers. In the Body section you can click on Body-Data and upload the picture you're interested in. Click in send and that'll try to make the API request. If it's successful, you'll get a 200 response and a URL back from the server, meaning you've uploaded it successfully. If you check it in Postman, you can see the picture.

Here's what the request looks like:

```
POST /v1/images/upload HTTP/1.1
Host: api.thecatapi.com
x-api-key: API_KEY
Content-Length: 232
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary7MA4YWxkTrZu0gW

------WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="file"; filename="/C:/Users/USER/Downloads/cat1.jpg"
Content-Type: image/jpeg

(data)
------WebKitFormBoundary7MA4YWxkTrZu0gW--
```

1. Host: header specifies hostname of the server that the client is trying to connect to.
2. Content-Type: request is uploading an image file named "cat1.jpg" using a type of data called "multipart/form-data". The image is in JPEG format and its content is included in the request body.
3. X-API-key: this header provides API key for authentication purposes
4. Content-Length: this header specifies the length of the request body in bytes. The value of this field is 232.

After the server received the request, it will parse the request body and use it to create a new entry in the Cat API database. The server will then return a response that includes information about the new entry such as the image URL and the database ID.

What is an HTTP Response

Here an example:

```
HTTP/1.1 200 OK
Date: Sun, 28 Mar 2023 10:15:00 GMT
Content-Type: application/json
Server: Apache/2.4.39 (Unix) OpenSSL/1.1.1c PHP/7.3.6
Content-Length: 1024

{
    "name": "John Doe",
    "email": "johndoe@example.com",
    "age": 30,
    "address": {
        "street": "123 Main St",
        "city": "Anytown",
        "state": "CA",
        "zip": "12345"
    }
}
```

The HTTP response is the message that a server sends back to a client in response to an HTTP request. It usually consists of a status line, headers, and a message body.

Status line contains HTTP version, status code indicating the outcome of the request and a corresponding message.

Headers provide additional information about the response, such as the content type of the message body or the date and time that the response was sent.

Message body contains actual response data, such as HTML, JSON or XML content.

Status Codes:

Code	Meaning
100	Continue
101	Switching Protocols
200	OK

Code	Meaning
201	Created
202	Accepted
203	Non-Authoritative Information
404	Not Found : The requested resource was not found on the server.
500	Internal Server Error : The server encountered an error while processing the request.
301	Moved Permanently: The requested resource has been permanently moved to a new URL.

Example of a JSON response from Cat API:


```
{
    "id": "ErDd1JRRT",
    "url": "https://cdn2.thecatapi.com/images/ErDd1JRRT.jpg",
    "width": 4282,
    "height": 6424,
    "original_filename": "cat1.jpg",
    "pending": 0,
    "approved": 1
}
```


This is a JSON response from Cat API request that appears to provide metadata about an image that has been uploaded or retrieved from the API.

Here’s what each thing means:

- id: A unique identifier for the image.
- url: The URL where the image can be accessed.
- width: The width of the image in pixels.
- height: The height of the image in pixels.
- original_filename: The original name of the file that was uploaded.
- pending: A flag indicating whether the image is still being processed by the API.
- approved: A flag indicating whether the image has been approved for public use by the API.